

Project Documentation: Desktop Computer Assembly and Verification

Project Title

Desktop Computer Assembly and Verification

Project Type

Hands-on IT hardware lab / desktop support project

Project Summary

For this project, I assembled a desktop computer and verified that it worked after putting the parts back together. This was a hands-on hardware lab where I worked with the main internal parts of a desktop, including the motherboard, CPU, CPU cooler, RAM, storage, power supply, and internal cables.

System / Device

Desktop computer

Small form factor desktop

Main parts handled: motherboard, CPU, CPU cooler, RAM, storage drive, power supply, internal cables, monitor, keyboard, mouse, and power cable

Objective

The goal of this project was to assemble a desktop computer from disassembled parts and make sure the system returned to a working state after assembly.

Tools and Materials Used

PC repair toolkit + Anti-static protection + Monitor + Keyboard + Mouse + Power cable + Smartphone camera for documentation + Desktop computer parts and screws

Work Performed

1. I started by setting up the workspace and making sure the parts, screws, and tools were available. I also followed anti-static precautions because I was handling internal computer components.
2. I installed the CPU onto the motherboard and made sure it was lined up correctly before locking it into place. I avoided touching the pins or conductive areas to make sure I didn't damage anything.
3. After that, I worked on the CPU cooler. I cleaned the contact area, and we didn't have thermal paste but I would have added some if we did. Also I mounted the cooler and connected the cooler cable to the CPU fan header on the motherboard.
4. Next, I installed the RAM. The system had 2 RAM modules, so I lined them up with the notches in the

slots and pressed them down until they were fully seated and locked in place. I made sure they were in the appropriate slots to enable Dual Channel mode in the BIOS.

5. Once the main parts were installed on the motherboard, I placed the motherboard into the desktop case. I lined it up with the case openings and screw holes, then secured it with the correct screws.
6. I also worked on the front-panel connections, including the power button, LED indicators, USB, and audio connections.
7. I installed and connected the storage device, then connected the needed power and data cables. I checked the inside of the case before powering it on to make sure the hardware was seated correctly and the cables were not blocking anything important.
8. After everything was connected, I plugged in the monitor, keyboard, mouse, and power cable. Then I powered on the computer to verify that the system worked after being assembled.

Skills Practiced

Desktop assembly + CPU installation + CPU cooler installation + RAM installation + Motherboard installation + Power supply connection + Storage installation + Internal cable connection + Front-panel connector setup + ESD-safe handling + Hardware verification + Technical documentation

Troubleshooting / Quality Checks

- Checked that the CPU was installed in the correct orientation.
- Made sure the CPU cooler was mounted securely and connected the CPU fan header. Verified the RAM was fully seated in place.
- Checked motherboard alignment before tightening screws.
- Reviewed cable connections before powering on the system.
- Verified the computer powered on after assembly.

Result

The desktop computer was assembled and checked after the hardware was installed. This project helped me get more comfortable with desktop hardware, motherboard layout, internal cabling, RAM installation, CPU cooler installation, and basic post-assembly verification.

- ❖ Made sure the CPU cooler was mounted firmly and connected to the CPU fan header. Verified the RAM was fully seated and locked into place.